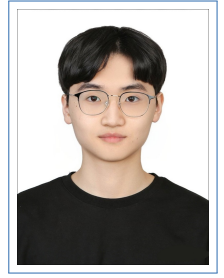


Song Woo Kim, MS Student

Curriculum Vitae

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Current Position

Feb. 2024 **MS**, Computer Science, Yonsei University
– present

Education

Feb. 2024 **BE**, Dept. of Computer Engineering, Yonsei University
Feb. 2022 **AD**, Dept. of Computer Engineering, Dongyang-miare University

Research Areas and Expertise

- Optimization Theory
- Robot Modeling and Control
- Robotic Path Planning
- Reinforcement Learning
- Imitation Learning

Selected Recent Publications

Journal articles

- 1 **S. W. Kim**, C. J. Park, J. W. Lee, S. M. Hyun, H. J. Jung and H. U. Yoon, "Robot Manipulator Planning using a Multiplicative Value Function for Dynamic Obstacle Avoidance," in *Proceedings of Korean Institute of Intelligent Systems*, Apr. 2024.
- 2 **S. W. Kim**, C. J. Park, J. W. Lee, S. M. Hyun, H. J. Jung and H. U. Yoon, "Design a Robot Manipulator Simulation Testbed for Various Path Planners," in *Proceedings of Korean Institute of Intelligent Systems*, Nov. 2023.
- 3 J. H. Kim, H. J. Jung, J. H. Yoo, D. H. Sim, **S. W. Kim** and H. U. Yoon, "An Approach to Design a Biomechanically-Inspired Reward Function to Solve a Patience Cube under Reinforcement Learning Framework," *International Conference on Intelligent Robots and Systems*, Sep. 2023.
- 4 **S. W. Kim**, D. H. Sim, J. Kim, J. H. Yoo, H. J. Jung, and H. U. Yoon, "Three Finger Robotic Platform for Demonstrating Dexterous Manipulations," in *Proceedings of Korean Institute of Intelligent Systems*, Apr. 2023.
- 5 D. H. Sim, **S. W. Kim**, and H. U. Yoon, "'Virtual Environment Design for Learning from Demonstration to Solve Patience Cube,'" in *Proceedings of Korean Institute of Intelligent Systems*, Apr. 2023.
- 6 J. H. Yoo, D. H. Sim, J. Kim, H. J. Jung, **S. W. Kim** and H. U. Yoon, "'Solving the C-shaped ring type Patience Cube as Planning Problem,'" in *Proceedings of Korean Institute of Intelligent Systems*, Apr. 2023.

Project-Based Research Keywords

- Learning from Demonstration

- Design a simulation environment for robotic task
- Visuo-motor task

Members

Fall 2022– **IEEE Robotics and Automation Society**

Honor and Awards

Academic

- Apr. 2023 **Best Paper Award**, *Korean Institute of Intelligent Systems*, Jeju, South Korea
Title: Virtual Environment Design for Learning from Demonstration to Solve Patience Cube
- Apr. 2023 **Best Paper Award**, *Korean Institute of Intelligent Systems*, Jeju, South Korea
Title: Three Finger Robotic Platform for Demonstrating Dexterous Manipulations
- Apr. 2023 **Best Paper Award**, *Korean Institute of Intelligent Systems*, Jeju, South Korea
Title: Solving the C-shaped ring type Patience Cube as Planning Problem

Programming Skill

- C/C++/C#/Java/Python
- MATLAB/Mathematica
- Simulation(Mujoco/Unity3D/Omniverse Isaac Simulation/Sapien)
- CAD(Fusion360)
- ROS1/ROS2
- Robotic Framework(Stable-baseline/Issac Lab/mani-skill/RSL-RL)
- Expert in developing application programs both Window and Linux environment